

Fig. 3: Actual-size PCB pattern of the transmitter unit

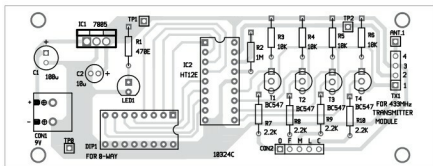


Fig. 4: Component layout of the PCB shown in Fig. 3

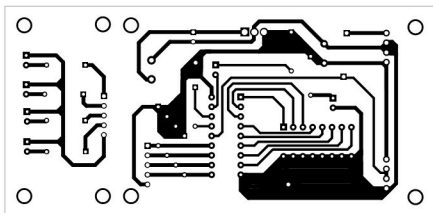


Fig. 5: Actual-size PCB pattern of the receiver unit

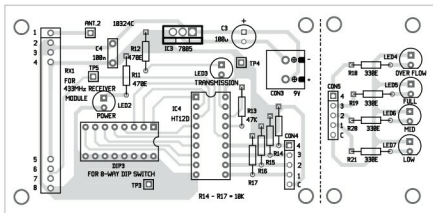


Fig. 6: Component layout of the PCB shown in Fig. 5

Test Points

Test point	Details
TP0, TP3	0V (GND)
TP1, TP4	5V
TP2, TP5	Pulses

transistors BC547 (T1-T4), connector CON2 for connecting the sensors in the overhead tank and a few other components.

Bases of T1 through T4 are connected with resistors R7 through R10. Each water-level sensor is connected to the circuit through R7 through R10. Water levels are marked as low (L), middle (M), full (F) and overflow (O) in the circuit. The common terminal is marked as C on CON2.

When water level increases, the transistors conduct and the output signal at their collector changes. Output of each transistor is fed to IC2. Encoded data from IC2 is transmitted through TX1.

Circuit diagram of the receiver unit is shown in Fig. 2. It consists of 433MHz RF receiver module RX1, a decoder HT12D (IC4), 5V regulator 7805 (IC3), DIP2 switch, LEDs (LED4 through LED7) and a few other components.

Data is received by RX1 and decoded by IC4. Different water levels are indicated by the glowing of LED4 through LED7. Resistors R14 through R17 are pullups connected to IC4.

General-purpose 9V batteries are used to power both the circuits. LED1 and LED2 are power indicators and LED3 is used to indicate a valid transmission status.

Construction and testing

An actual-size, single-side PCB for the transmitter unit is shown in Fig. 3 and its component layout in Fig. 4. An actual-size, single-side PCB for the receiver unit is shown in Fig. 5 and its component layout in Fig. 6. After assembling the circuits on their PCBs, enclose these